BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

LESSON 072 · File Handling

Name: _____ Class: _____ Date: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. What is a parameter?

Q2. How does phishing trick people?
A. Tries every possible password
B. Floods a server with traffic
C. Repairs damaged files
D. Pretends to be trusted

Q3. “Fibre carries electrical signals through copper wire.” Correct this. Explain why.

Q4. Desktops and moving tablets need network links. “Wi-Fi needs a cable from each tablet to the switch.” Correct this. Which links suit these devices?

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I Do · Read new knowledge

Key vocabulary

Open file — open(filename) returns a handle to an existing text file.

Read line — handle.readLine() obtains the next line of text from an opened file.

Write line — handle.writeLine(text) adds a line to an opened file.

Close file — handle.close() releases the file when finished.

Algorithm — A defined set of processing instructions.

Testing — Checking expected outcomes against actual program behaviour.

New learning

Open file: open(filename) returns a handle to an existing text file. In the daily attendance file, identify where this technique applies.

Read line: handle.readLine() obtains the next line of text from an opened file. In the revision notes file, identify where this technique applies.

Write line: handle.writeLine(text) adds a line to an opened file. In the library loans file, identify where this technique applies.

Close file: handle.close() releases the file when finished. In the temperature log, identify where this technique applies.

Worked example

A daily attendance file uses open file and read line. Explain what each contributes. [4 marks]

1. Identify the program's purpose and the pieces of data needed.

2. Open file [1]; open(filename) returns a handle to an existing text file. [1].

3. Read line [1]; handle.readLine() obtains the next line of text from an opened file. [1].

4. Each feature needs a correct technical fact plus an explanation.

We Do · Practise with guidance

Guided 1

A revision notes file needs open file. Explain the purpose. [2 marks]

Worked method: Open file [1]; open(filename) returns a handle to an existing text file. [1].

Guided 2

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Explain read line for a library loans file. [2 marks]

Worked method: Read line [1]; handle.readLine() obtains the next line of text from an opened file. [1].

Guided 3

When programming a temperature log, explain why write line and close file matter. [4 marks]

Worked method: Write line [1]; handle.writeLine(text) adds a line to an opened file. [1]. Close file [1]; handle.close() releases the file when finished. [1].

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You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Code First · Predict → Write → Test → Improve

Complete the statement that appends "Backup complete" to an existing text file.

```
myFile = open("status.txt")
// write a line
myFile.close()
```

write

Question 1

State TWO relevant ideas or operations from this topic. [2 marks] Source: Original OCR J277-style teaching question; not an official past paper.

Question 2

Explain read line and write line in a sport-result text file. [4 marks] Source: Original OCR J277-style teaching question; not an official past paper.

Question 3

A programmer makes a teacher diary file. Explain the roles of open file, write line and close file. [6 marks] Source: Original OCR J277-style teaching question; not an official past paper.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

Exit Ticket · Answer before checking

open(filename) returns a handle to an existing text file.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. A named input received by a subprogram when called with an argument.

Q2. D — Pretends to be trusted

Q3. Fibre uses light. It supports fast data transfer over long distances and resists electromagnetic interference. Installation can cost more.

Q4. Ethernet uses cables. Wi-Fi uses radio, so tablets can move around, but interference can affect it. Choose links to suit the distance and data rate needed.

I Do / We Do pages · Guided practice

Q1. Open file [1]; open(filename) returns a handle to an existing text file. [1].

Q2. Read line [1]; handle.readLine() obtains the next line of text from an opened file. [1].

Q3. Write line [1]; handle.writeLine(text) adds a line to an opened file. [1]. Close file [1]; handle.close() releases the file when finished. [1].

You Do pages · Code First

```
myFile = open("status.txt")
myFile.writeLine("Backup complete")
myFile.close()
```

Expected result: No screen output; "Backup complete" appended to status.txt

You Do pages · Independent questions

Q1. Open file [1]; Read line [1].

Q2. Read line [1]; handle.readLine() obtains the next line of text from an opened file. [1]. Write line [1]; handle.writeLine(text) adds a line to an opened file. [1].

Q3. Open file [1]; open(filename) returns a handle to an existing text file. [1]. Write line [1]; handle.writeLine(text) adds a line to an opened file. [1]. Close file [1]; handle.close() releases the file when finished. [1].

Exit Ticket · Model answer

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

True

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question:

2. Write a complete new answer in GREEN PEN. Show working if needed: